



The AI Venture Operating System

*A Technical White Paper on the Architecture, Financial Engines,
Quantitative Methods, and Workflows of the Anker Platform*

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A note on data integrity. Every figure attributed to “the platform” in this paper is drawn from the live production database (Neon Postgres) at the time of writing, or from the source code. Where a number is a *target* rather than an achieved result, it is labelled (*target*). Testimonial blocks are left as clearly-marked placeholders to be filled with verbatim, attributable quotes; no quotation in this paper should be treated as genuine until a named source has confirmed it in writing. Platform screenshots are collected as **Figures 1–14** in the companion *Figure Supplement* ([anker-whitepaper-figures.pdf](#)), referenced inline at the relevant points.

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1 Abstract

Anker is a multi-persona **venture operating system**: a single web application that a startup **founder** uses to raise a round, a **venture / fund general partner (GP)** uses to source deals and run the back office, and a **limited partner (LP)** uses to monitor their capital — each persona seeing a surface scoped to their role but drawing on one shared data and intelligence layer.

Unlike a marketing site with a login, Anker embeds **institutional-grade financial machinery** directly in the product: an option-pricing-model (OPM) 409A valuation engine, a private-credit loan-servicing engine with covenant tracking, an equity share-plan engine with cliff-aware vesting, and a fund-administration layer that computes the standard private-capital performance metrics (TVPI, DPI, RVPI, MOIC, and money-weighted IRR) from a live ledger. Around this sits an **AI layer** — a provider-agnostic model router feeding a tool-calling agent that adapts into three distinct *persona copilots* — and a **semantic matching** subsystem over a database of **47,275** investors and decision-makers.

This paper documents the system end-to-end: the architecture, the exact mathematics behind each engine (with derivations), the machine-learning and quantitative methods, the user workflows, and the security posture. It closes with an honest empirical section grounded in the real production dataset.

2 Introduction & Market Position

2.1 The problem

The venture lifecycle is served by a patchwork of point tools: one product for the cap table, another for the data room, a CRM for investor relationships, a spreadsheet for the fund model, a separate portal for LP reporting, and nothing that connects the founder raising capital to the investor deploying it. The data that matters is fragmented across these tools and across the boundary between the two sides of the table.

Anker’s thesis is that **the founder side and the fund side are the same graph** — companies, investors, rounds, positions, and people — and that a single system holding that graph can (a) remove the integration tax, and (b) apply intelligence across the whole lifecycle rather than within a single silo.

▷ See **Figure 1** in the companion Figure Supplement (Anker platform screenshots).

Caption: Anker marketing home in light mode — editorial serif hero (“Your venture ready to build”) over a generative particle field; primary CTAs *Start Fundraising / See How It Works*; trust bar reading 40K+ HNWI & LPs, 50+ countries, 60K+ investors in database.

2.2 Positioning

Anker positions itself honestly as **pre-seed-stage software building toward Carta-class parity** across three suites:

Suite	Audience	What it does
Anker Founder Suite	Founders	Find investors, model the round, share the data room
Anker Fund OS	VCs / GPs	Source deals, run fund admin, report to LPs
Anker Investor Room	LPs	Self-serve capital account, distributions, documents

The three suites are not three products; they are three *views* of one graph, selected by the viewer's *persona*.

▷ See **Figure 2** in the companion Figure Supplement (Anker platform screenshots).

Caption: The Products mega-menu — a left rail of suite tabs (Founder Suite / Fund OS / Investor Room) driving a right panel of features.

2.3 Scale of the underlying dataset

The investor-discovery surface is backed by a substantial real dataset.

▷ See **Figure 3** in the companion Figure Supplement (Anker platform screenshots).

Caption: The public *Investor Database* page — 18,000+ firms, 47,000+ decision-makers, 84 geographies, weekly re-crawl. These are not placeholders: the production `investors` table holds **47,275** rows.

3 System Architecture

3.1 Stack

Anker is a server-first **Next.js (App Router)** application in **TypeScript**, styled with **Tailwind CSS v4**, deployed on **Vercel**, backed by **Neon** (serverless Postgres), with authentication delegated to **Supabase Auth**. The package manager is **pnpm**. The AI layer routes across multiple LLM back-ends.

3.2 The pure-math / service split

A deliberate decision underpins every financial engine: the **mathematics is isolated in pure, dependency-free modules** (`lib/modules/*`) that import no database driver. This yields three properties: (1) *testability* — every formula is exercised against textbook expected outputs; (2) *isomorphic execution* — the identical function runs on the server (authoritative computation) and in the browser (live “as-you-type” projections), guaranteeing the number shown equals the number stored; (3) *auditability* — because the math never touches I/O, a reviewer can verify a valuation from a single file.

3.3 The derived-value discipline

Quantities that can be computed are never hand-typed. A loan's outstanding balance is $\text{principal} - \sum \text{principal_portion}$ over the payment ledger; a grant's exercised total is $\sum \text{quantity}$ over the exercise ledger; a fund's called capital is the sum of capital-call line items. Each is recomputed from its ledger after every mutation, eliminating an entire class of reconciliation bugs endemic to spreadsheet-based management.

4 Identity, Personas & Access Control

Every destination is tagged with the personas that may see it: `personas: ["founder"|"vc"|"lp"]`. A user resolves to an active persona via workspace membership; a user with no membership, or a **platform owner**, resolves to `null` (“sees everything”).

Two *platform tiers* sit above per-workspace roles: **owner** (full oversight, firewalled from tenant private records) and **admin** (staff), with $\text{owner} \supset \text{admin}$. Route protection is centralized in a server guard that fails safe and always lets owners through. Registration is **invite-only and fails closed**: without a configured `SIGNUP_INVITE_CODE` secret, nobody can register. Login returns a deliberately generic “Invalid email or password” (no user enumeration) and redirects only to internal paths (open-redirect guard; see §12).

5 Application Shell & Information Architecture

The in-app shell is the logged-in continuation of the marketing site. A single **navigation taxonomy** (`lib/nav/taxonomy.ts`) is the source of truth for both surfaces, so suite names, wording, and grouping are identical on both sides of the login. The shell provides a website-style top bar with a persona-scoped Products mega-menu, a contextual left rail carrying the full grouped destination set (~ 28 for a GP), a command palette (`Cmd+K`) that derives its entries from the taxonomy and searches by description, and a mobile bottom-tab + sheet treatment. Page chrome follows the site’s editorial system: serif display titles, mono uppercase eyebrows with a square accent bullet, muted body.

▷ See **Figure 4** in the companion Figure Supplement (Anker platform screenshots).

Caption: The in-app shell (Fund OS persona) — top bar, contextual rail of ~ 28 grouped destinations, and the workspace.

6 The Tier-1 Financial Engines

Each ships as a pure-math module verified against textbook results, a server service that persists and scopes, a REST surface, and a detail UI that projects live.

6.1 The 409A / Option-Pricing-Model Valuation Engine

A 409A valuation sets the fair market value (FMV) of *common* stock — the basis for option strike prices — typically below the preferred price, because common lacks the preferences of preferred. Anker uses the **Option Pricing Model (OPM)** back-solve, treating each class of equity as a **call option** on total equity value.

6.1.1 Black–Scholes as the primitive

The OPM prices equity claims as European calls on total equity value S struck at each *break-point*:

$$C(S, K, \sigma, r, T) = S \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad (1)$$

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (2)$$

where S is total equity value, K a breakpoint, σ equity volatility, r the risk-free rate, T years to liquidity, and Φ the standard-normal CDF, implemented with the **Abramowitz–Stegun** rational approximation (max abs error $\approx 7.5 \times 10^{-8}$). Boundary tests: $C(S, 0) = S$ and $C(0, K) = 0$.

6.1.2 Two-class waterfall & breakpoints

With one preferred series (liquidation preference LP , fully-diluted ownership fraction p) and common, total value W is allocated across breakpoints: $[0, LP)$ preferred only; $[LP, CP)$ common only; $[CP, \infty)$ pro-rata, where the *conversion point* is $CP = LP/p$. Each interval's value to a class is a *call spread* (a difference of Black–Scholes calls). The common allocation:

$$V_{\text{common}}(W) = [C(W, LP) - C(W, CP)] + (1 - p) C(W, CP). \quad (3)$$

6.1.3 The back-solve

The observed preferred price implies total equity value. Anker solves for the W such that OPM-allocated preferred value equals $\text{prefShares} \times \text{recentPrice}$,

$$\text{find } W \text{ s.t. } V_{\text{preferred}}(W) = \text{prefShares} \times \text{recentPrice}, \quad (4)$$

by **bisection** (robust; monotone objective; derivative-free). The per-share common FMV is $V_{\text{common}}(W^*)/\text{commonShares}$, optionally reduced by a discount for lack of marketability (DLOM).

6.1.4 Worked verification

For **8,000,000** common and **2,000,000** preferred at a recent preferred price of **\$1.00** ($\sigma = 0.6$, $r = 0.04$, $T = 4$, DLOM off), the back-solve recovers a total equity value at which preferred is worth exactly **\$2,000,000**, and common FMV solves to $\approx$ **\$0.4441** — i.e. common is marked at $\approx 44\%$ of the preferred price. This is the module's canonical regression test.

Caveat (from the product's own methodology writeup): the two-class OPM is appropriate for a single non-participating preferred series. Multi-series stacks, participating preferred, and safe-harbor reliance require a fuller waterfall and an independent appraisal.

▷ See **Figure 5** in the companion *Figure Supplement* (Anker platform screenshots).

Caption: The 409A engine — common FMV updates live as OPM inputs are typed; a breakpoint-allocation table; a methodology writeup; a downloadable board-approval report.

6.2 The Loan-Servicing (Private-Credit) Engine

6.2.1 Amortization

For a fully-amortizing loan, the **level payment** A on principal P , periodic rate $i = \text{APR}/f$, over n periods, is the annuity formula

$$A = P \cdot \frac{i(1+i)^n}{(1+i)^n - 1}. \quad (5)$$

Each period splits into $\text{interest}_t = i \cdot \text{balance}_{t-1}$ and $\text{principal}_t = A - \text{interest}_t$, with $\text{balance}_t = \text{balance}_{t-1} - \text{principal}_t$. Interest-only, bullet, and revolving structures are also supported. *Verification:* \$100,000 at 12% APR over 12 monthly periods yields a level payment of **\$8,884.88**.

6.2.2 Accrual & the interest-first ledger

Between scheduled payments, interest accrues on an **Actual/365** basis:

$$\text{accrued} = \text{balance} \times \text{APR} \times \frac{\Delta_{\text{days}}}{365}. \quad (6)$$

A recorded payment is applied *interest-first*, the remainder to principal, and the outstanding balance is recomputed from the ledger. Covenants (financial, reporting, affirmative, negative) are tracked with status and tested-at date.

▷ See **Figure 6** in the companion Figure Supplement (Anker platform screenshots).

Caption: Loan servicing — summary tiles, repayment-progress bar, editable terms that re-project the amortization schedule, the payment ledger, the covenant tracker.

6.3 The Share-Plans (Equity Vesting) Engine

6.3.1 Vesting mathematics

For a grant of N options, total vest M months, cliff c months, and m whole months of credited service:

$$\text{vested}(m) = \begin{cases} 0, & m < c, \\ \min\left(N, \left\lfloor N \cdot \frac{\min(m, M)}{M} \right\rfloor\right), & m \geq c. \end{cases} \quad (7)$$

Nothing vests before the cliff; at the cliff the elapsed fraction vests *at once* (12-of-48 $\Rightarrow$ 25%); thereafter straight-line to 100% at M . Termination freezes credited service; exercisable = $\max(0, \text{vested} - \text{exercised})$. *Verification:* 48,000 options, 48-month vest, 12-month cliff: 0 at month 11; **12,000** at month 12; **24,000** at month 24; capped at **48,000** past month 48; termination at month 30 freezes at **30,000**.

▷ See **Figure 7** in the companion Figure Supplement (Anker platform screenshots).

Caption: Grant detail — vesting/exercise tiles and progress bars, editable terms re-projecting a 48-row schedule with the cliff row highlighted, exercise ledger, cap-table contribution.

7 Fund OS — Fund Administration & Performance Mathematics

7.1 The performance metric suite

$$\text{DPI} = \frac{\text{Distributions}}{\text{Paid-in}}, \quad \text{RVPI} = \frac{\text{NAV}}{\text{Paid-in}}, \quad \text{TVPI} = \text{DPI} + \text{RVPI} = \frac{\text{Distributions} + \text{NAV}}{\text{Paid-in}}. \quad (8)$$

Gross MOIC on a position is (realized + unrealized)/cost.

7.2 Money-weighted return (IRR)

The fund IRR is the rate x solving the NPV identity over dated cash flows CF_k at times t_k (years),

$$\sum_k \frac{CF_k}{(1+x)^{t_k}} = 0, \quad (9)$$

calls negative, distributions plus terminal NAV positive; solved numerically (Newton–Raphson with a bisection fallback for irregular flows). NAV of record is the sum of position fair values from the Valuations engine.

7.3 Waterfall, pacing & forecasting

Carried-interest distributions follow the fund’s waterfall (return of capital → preferred return / hurdle → GP catch-up → carry split), modelled so LP/GP splits reconcile to the ledger. Fund Forecasting projects deployment pacing, reserve policy, and projected returns — the portfolio-construction analogue of a position-sizing strategy.

7.4 The Fund OS module map

Fund OS is not a single dashboard but a workspace of ~24 dedicated modules, each a route under `/dashboard/portfolio/fund/*`, all reconciling to the same ledger:

- **Capital & cash flow** — `calls` (capital calls, with a per-call detail view, a table view, and a “new call” wizard), `distributions` (waterfall-driven, per-distribution detail + table + new), `ledger` (the append-only source of the derived NAV / paid-in / DPI).
- **Investors & economics** — `partners` (GP/LP roster), `lps/[lpId]/statement` (per-LP capital-account statement), `lp-imports` (bulk onboarding), `economics` (management fee, carry, hurdle), `syndication` (SPVs / co-invest).
- **Portfolio & deals** — `deals` (pipeline, with a `[dealId]` detail), `investments` (positions of record), `explorer`, `performance` (TVPI/DPI/RVPI/IRR), `tear-sheet`, `assessment` (fund-strength framework, incl. an assessment wheel).
- **Governance & reporting** — `documents`, `legal` (review, fields, per-document `[docKey]`), `reports`, `management`, `plan`, plus `fund-tax` and the standalone `tools/vc-fund-model` and `tools/fund-of-funds` modellers.

Every dollar figure on every one of these pages is *derived* from the ledger (§3.3), not hand-entered — the single discipline that lets two-dozen views stay mutually consistent.

▷ See **Figure 8** in the companion *Figure Supplement* (Anker platform screenshots).

Caption: Fund OS overview — fund profile, “Fund workspace” card grid, rollup tiles, LP subscription funnel.

8 The AI System — Model Router, Persona Agents & Tooling

8.1 Provider-agnostic model router

A **model router** abstracts across providers (Anthropic Claude, and via the DashScope gateway Qwen 3.x, GLM, DeepSeek, Kimi, and others), selecting a model per task tier with automatic fallback.

8.2 The tool-calling agent loop

The assistant is an **autonomous agent** reasoning in steps, emitting one JSON action per step over a catalog of platform-integrated tools (CRM, deal pipeline, warm-intro paths, outreach,

LP matchmaking, thesis scoring, firm enrichment, web search/crawl, document/spreadsheet generation), batching work to respect quotas and pre-processing uploads (PDF text, image OCR/vision).

8.3 Persona agents

The assistant adapts into three agents by active persona, each with its own role, tool scope, and integrated features derived from the shared taxonomy:

Agent	Persona	Bias
Founder Copilot	founder	Investor discovery, warm intros, pitch/data-room, dilution, runway
Fund Copilot	vc	Deal sourcing/IC, thesis scoring, LP matchmaking, monitoring, back office
Investor Copilot	lp	Capital accounts, call/distribution explanations, documents, performance

The active persona is injected into the system prompt with the persona’s real destinations; the UI surfaces persona-specific suggested tasks. Owners default to the Fund Copilot.

8.4 AI workflows — current and expanding

Beyond conversational Q&A, the agent runs *multi-step operational workflows* end-to-end. The ones in production today:

- **Founder discovery run** — from a thesis or uploaded deck, embed → retrieve → score → dedupe → emit a ranked, exportable investor list with match rationales (the same engine the campaign orchestrator calls, §10).
- **Firm enrichment & thesis scoring** — fill thin investor records from public sources and grade a batch of firms against a thesis (≤ 40 /call), returning a spreadsheet rather than an opaque score.
- **Deck / data-room ingestion** — PDF text extraction plus image OCR/vision to build a structured startup profile used downstream by assessment and matching.
- **Outreach drafting** — per-recipient personalized email bodies with one-click interest links (§10), generated in concurrency-limited batches.
- **Reply handling** — inbound replies are parsed and classified (`lib/ai/reply-handler.ts`) to advance the CRM stage and surface hot leads in the shared inbox.

Expanding: document/spreadsheet co-authoring, a Fund-Copilot IC-memo generator over the deal record, and an Investor-Copilot capital-account explainer that narrates calls and distributions in plain language — each a thin agent layer over data and engines that already exist, which is why they can ship incrementally.

▷ See **Figure 9** in the companion Figure Supplement (Anker platform screenshots).

Caption: The AI Assistant rendered as the Fund Copilot — eyebrow “AI Agent · Fund Copilot”, serif title, persona-specific suggestion chips.

9 Quantitative & Machine-Learning Methods

This section documents the methods actually present. It deliberately does **not** report market-trading backtests for a strategy the platform does not run; the quantitative core is derivatives-style valuation, credit modelling, portfolio construction, and semantic ranking.

9.1 Semantic investor matching

Discovery ranks **47,275** investors for a company/thesis via **embedding-based semantic retrieval**. A company/thesis and each investor profile are encoded to dense vectors $u, v \in \mathbb{R}^d$; fit is scored by **cosine similarity**

$$\text{sim}(u, v) = \frac{u \cdot v}{\|u\| \|v\|}, \quad (10)$$

computed in Postgres via pgvector’s distance operator (HNSW-indexed), then combined with hard filters (stage, geography, check size, sector, recent activity) and *warm-intro* signals from a shortest-path over the relationship graph, so the final ranking blends *fit* and *reachability*.

9.2 Thesis scoring & enrichment

A scoring tool grades firms against a thesis (bounded batches, ≤ 40 per call); an enrichment tool fills thin records from public sources. Both are LLM-in-the-loop with rate-limited failover, producing exportable spreadsheets rather than opaque scores.

9.3 Option-pricing math as the derivatives core

The 409A engine (§6.1) *is* the platform’s derivatives-pricing model: it treats each equity class as a call option and prices call spreads across breakpoints with Black–Scholes — the same mathematics used to price options in public markets, applied to private structures, and legally consequential (it sets strike prices).

9.4 On “trading strategies”

Anker is a venture operating system, not a trading system, and this paper will not manufacture a strategy or backtest to fit a narrative. The honest mapping: option pricing (§6.1) is the derivatives math; credit servicing (§6.2) the fixed-income math; pacing/reserves (§7.3) the portfolio-construction math; IRR/TVPI/DPI (§7) the performance math; semantic ranking (§9.1) the ML. Any future market-facing strategy would be documented here with the same rigour — and never before it exists.

10 Core Workflows & the Founder Campaign Engine

10.1 Founder: from “Pitch us” to investor inboxes

The founder’s headline workflow is model (Cap Table, 409A, Share Plans, Runway) → Raise Pipeline (by stage) → Data Room (12-section diligence taxonomy, watermarked / tokenized / expiring share links with view analytics) → Outreach. The outreach step is not a mail-merge; it is an autonomous **campaign engine** that turns a single “Pitch us” submission into a ready-to-send, personalized investor campaign.

10.1.1 The orchestrator pipeline

One submission is driven through an idempotent, status-gated state machine (`lib/campaign/orchestrator.ts`). Each stage flips the submission’s status so an overlapping cron tick can never double-process it:

$$\text{received} \xrightarrow{\text{claim}} \text{assessing} \longrightarrow \begin{cases} \text{declined} & (\text{gate fails} \rightarrow \text{feedback}) \\ \text{campaign_ready} & (\text{gate passes} \rightarrow \text{matched+drafted}) \end{cases}$$

1. **Ingest** — pull deck + up to three data-room docs from Blob storage.
2. **Extract** — OCR/vision + text to a structured **StartupProfile**; founder-typed form fields are treated as *higher-trust* than extraction and overlay it, so a deck that fails to parse still yields a usable profile.
3. **Readiness gate** — a conservative LLM assessment scores the startup against a five-dimension rubric; below the threshold (default **62/100**) the submission is auto-declined *with constructive feedback*, and the pipeline stops. This is the “100+ startups assessed” capability (§13.3) in operation.
4. **Match** — `runFounderMatching` ranks the investor universe; only contacts with an email and a match score $\geq$ the floor (default **55**) survive, capped at `maxInvestors` (default **100**).
5. **Create campaign + mint tokens** — an `outreach_campaign` row plus one `campaign_crm_entry` per investor; each entry gets a signed *interest token* backing one-click *Interested* / *Not-interested* / *View* links.
6. **Draft** — per-recipient subject + body generated in a concurrency-limited batch (8-wide), each grounded in the startup profile and the specific investor’s match rationale.
7. **Hand off** — status becomes `campaign_ready`; the sender takes over.

10.1.2 Wave-based sending

A cron sender (`/api/cron/campaign-send`, `CRON_SECRET`-gated, fail-closed) ships the next **wave** of drafted-but-unsent emails per active campaign (default wave **20**). Small waves protect domain reputation and let early results be observed before the next wave goes. Each message carries the founder story, a deck link, the one-click interest links, an Anker AI signature, and a reply-to/BCC so replies land in the shared inbox; when a campaign’s queue drains, the submission flips to **completed** and the founder gets a wrap-up email. Two admin toggles govern automation:

Setting	Default	Effect
<code>readinessThreshold</code>	62	Below this, auto-decline with feedback
<code>scoreFloor</code>	55	Minimum match score to include an investor
<code>maxInvestors</code>	100	Cap on investors per campaign
<code>waveSize</code>	20	Emails per send wave
<code>autoAssess</code>	on	Cron auto-processes new submissions
<code>autoSend</code>	on	When off, a human releases each campaign

Settings live in a single DB row read on every assessment/send, so an operator retunes the engine from the console with no redeploy.

10.1.3 Deliverability guardrails

Because the investor list is *organically sourced* rather than opt-in, list hygiene is enforced before send by a dependency-free verifier (`lib/outreach/email-quality.ts`) applying four ordered gates — (1) format regex, (2) role-address rejection (`info@`, `hello@`, `noreply@`, ...), (3) a hard-bounce suppression list carried forward from the prior batch, and (4) a live DNS MX check (per-domain cached) — so a message only sends to an address that is well-formed, personal, not previously bounced, and whose domain can actually receive mail. Post-send, a poller (`lib/email/resend-sync.ts`) reconciles Resend's `last_event` back into per-message status and open/click counters for the inbox. The measured effect of these guardrails is reported in §13.4.

10.2 VC / GP

Deal Flow → IC → close into positions; LP Matchmaking; the full back office (capital calls, distributions, SPVs / syndication, KYC/AML, fund tax, contracts, compliance) across the ~24 Fund OS modules (§7.4); reporting and monitoring. GPs also run their *own* outreach (a dedicated LP campaign builder, `/dashboard/outreach/lp-campaign`) over the same engine — e.g. the family-office webinar campaign in §13.4.

10.3 LP: the Investor Room

LPs get a self-serve portal (`/lp`) — a branded overview, a `documents` vault, a `distributions` view, and `help` — reading the same capital-account, call, and distribution records the GP maintains, with an explicit, badged owner-oversight mode (the one logged exception to the owner→tenant firewall, §12). The GP side mirrors it with bulk `lp-imports` and a per-LP `lps/[lpId]/statement` capital-account statement.

▷ See **Figure 10** in the companion *Figure Supplement* (Anker platform screenshots).

Caption: The LP Investor Room — branded header with owner-oversight badge, sub-nav, capital-account summary.

11 Billing, Credits & Metering

Anker meters usage through a **Stripe** integration (test-mode connected). The read path stays off Stripe: a webhook mirrors subscription state into local tables (`billing_customers`, `billing_subscriptions`), and an append-only **credit ledger** (`billing_credit_ledger`) records grants and consumption; the balance is the running sum. Three plans — Starter / Pro / Scale — carry monthly AI-credit allotments granted on each paid invoice. The Stripe client loads lazily and is gated on configuration.

12 Security, Compliance & Auditability

- **Authentication** — Supabase sessions; invite-only, fail-closed registration; generic login errors.
- **Open-redirect remediation** — the post-login `?next=` target was applied without validation, allowing `?next=https://evil.com` to redirect a freshly-authenticated user off-site (CWE-601). Both the login page and OAuth callback now accept internal paths only.

- **Owner→tenant firewall** — owners see aggregate/existence data, not tenant private records, enforced structurally; one explicit, logged LP-portal oversight exception.
- **Immutable audit log** — an append-only `audit_events` trail (actor, action, target, meta-data, IP, UA) with a filterable admin feed; writes are best-effort so auditing never breaks the action it records.
- **Compliance register** — enumerates the regulatory obligations a U.S. VC fund faces and flags what applies to the fund’s profile.

13 Evaluation & Empirical Grounding

This section reports **only** what the production system actually contains.

13.1 Dataset (production, at time of writing)

Asset	Count	Notes
Investors / decision-makers	47,275	Core discovery dataset (site markets “60k+”; see note)
— of which with an email	9,058	The addressable subset for outreach
Contacts	1,378	Relationship-graph nodes
Outreach campaigns	9	Founder + GP campaigns created by the engine
Campaign CRM entries	100	Matched-investor rows across campaigns
News articles (published)	88	Newsroom content
Funds	1	Summit Venture Studio — Fund II (canonical fund)
LPs on the fund	8	Capital accounts
Investment positions	8	NAV of record
Founder submissions assessed	6	Inbound “Pitch us” applications to date

13.2 Engine correctness (regression suite)

Engine	Test	Expected	Result
409A / OPM	8M common / 2M pref @ \$1	pref → \$2.00M; common ≈ \$0.4441	pass
Loan amortization	\$100k, 12% APR, 12 mo	level payment \$8,884.88	pass
Vesting	48k / 48mo / 12mo cliff @ 24mo	24,000; freeze@30mo ⇒ 30,000	pass
Billing ledger	+5,000 then −120	balance 4,880	pass

13.3 On “100+ startups assessed”

The production `founder_submissions` table currently holds **6** inbound applications. A claim of “100+ startups assessed” is therefore **not** supported by the platform’s own records and is presented here, if at all, only as a *target*. The assessment *capability* (a structured, multi-field fund-strength and company assessment framework) exists; the *volume* is what it is. A research paper that inflates its own evaluation set is not a research paper.

On “60k+ investors”. The marketing site advertises a 60k+ investor database. The production `investors` table holds **47,275** rows at the time of writing (§13.1), of which **9,058** carry an email and are therefore addressable for outreach. The 60k+ figure is a rounded, forward-looking marketing number; the audited count is 47,275.

13.4 Email deliverability & engagement (measured)

Two real outreach campaigns were run through the engine and metered by the email provider (Resend) on the `an-ker.de` sending domain. These are provider dashboard figures, not self-reported:

Metric	Family-office webinar	Founder→VC outreach
Campaign type	GP-run, curated	“Pitch us” + engine
Provider window	15-day	30-day (Aug 1–15)
Emails sent	419	300
Deliverability	84.25%	90.67%
Delivered (approx.)	353	272
Bounce rate	15.27% (64)	9.33% (28)
permanent / transient	42 / 22	24 / 4
Spam complaints	0% (0)	0% (0)
Open rate	58.64%	35.29%
Click rate	25.5%	10.29%

Reading the numbers honestly.

- **Zero spam complaints across 719 sends** — the strongest single signal that the targeting and content are legitimate, not spray-and-pray.
- **Deliverability rose 84.25% → 90.67% and bounce roughly halved (15.27% → 9.33%)** between the earlier curated batch and the later founder campaign — consistent with the pre-send verification pipeline and the carried hard-bounce suppression list (§10) doing their job. Permanent bounces (stale addresses) dominate both, which is exactly what an organically-sourced, non-opt-in list produces — and exactly what the suppression list removes going forward.
- **Both bounce rates still sit above the provider’s ~4% “healthy” line.** That is the honest cost of an organic database and the clearest near-term work item (§15): continued list cleaning until bounce is under control.
- **Engagement tracks intent, as it should.** The invited family-office webinar drew a 58.6% open / 25.5% click; cold founder→VC outreach drew 35.3% / 10.3%. Even the colder number is strong against a typical ~20% cold-outreach open-rate benchmark.

The provider dashboards behind these figures are reproduced in the companion Figure Supplement: the **family-office webinar** deliverability and engagement panels (**Figures 11–12**) and the **founder→VC outreach** panels (**Figures 13–14**).

▷ See **Figure 11** in the companion Figure Supplement (Anker platform screenshots).

Provenance & caveats. Figures are Resend domain-level dashboard metrics for the two windows shown. Open rates are pixel-based and both undercount (privacy proxies) and overcount (prefetch); treat them as directional. The per-message `outreach_messages` table recorded 285 provider sends, a subset of total domain traffic, and their historical provider IDs are past Resend’s GET-retention window — so the dashboard aggregate above, not the local table, is the authoritative source for these two campaigns.

14 Practitioner Commentary

These blocks are placeholders. None of the quotations below has been verified, and none should be published until a **named source has supplied and approved the exact wording in writing.** Fabricating a testimonial — especially one attributed to a real institution or an identifiable individual — would be misinformation and is out of scope. Replace each block with a verbatim, attributable quote and obtain written consent to be named.

On the Fund OS back office —

[Placeholder — verbatim quote to be supplied and approved by a named GP at a venture fund. Include: name, title, firm, date.]

On investor discovery & matching —

[Placeholder — verbatim quote to be supplied and approved by a named allocator or family-office principal. Include: name, title, entity, date.]

On the quantitative / valuation engines —

[Placeholder — verbatim quote to be supplied and approved by a named finance academic or practitioner. Include: name, title, institution, date; if the source has media appearances, cite the specific program and air date rather than paraphrasing.]

Method for collecting these: send the relevant section (e.g. §6.1 for the valuation engine) to the source, capture a one- to three-sentence on-the-record reaction verbatim, and record written consent. Only then does the quote belong in the paper.

15 Roadmap

- Multi-fund & multi-series — generalize the single canonical fund and extend the OPM to multi-series / participating-preferred waterfalls.
- Rate limiting & password policy on the auth routes.
- Deeper analytics — fundraising-funnel and portfolio-performance dimensions.
- Assessment volume — grow the corpus so a future evaluation reports results, not targets.
- Deliverability / list hygiene — drive bounce under the provider's ~4% line: continuous suppression-list growth, a paid verifier for the addressable subset, and per-domain send throttling (building on the measured 84.25%→90.67% gain in §13.4).
- Billing go-live — production webhook secret + environment; metered credits.

16 Appendix A — Formula Reference

Black–Scholes call. $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$, $d_1 = [\ln(S/K) + (r + \sigma^2/2)T]/(\sigma\sqrt{T})$, $d_2 = d_1 - \sigma\sqrt{T}$.

OPM common allocation. $V_{\text{common}}(W) = [C(W, LP) - C(W, CP)] + (1 - p)C(W, CP)$, $CP = LP/p$.

409A back-solve. find W^* s.t. $V_{\text{pref}}(W^*) = \text{prefShares} \cdot \text{recentPrice}$ (bisection); $\text{commonFMV} = V_{\text{common}}(W^*)/\text{commonShares} \cdot (1 - \text{DLOM})$.

Level loan payment. $A = P i(1+i)^n / [(1+i)^n - 1]$, $i = \text{APR}/f$.

Actual/365 accrual. $\text{accrued} = \text{balance} \cdot \text{APR} \cdot \Delta_{\text{days}}/365$.

Vesting. $\text{vested}(m) = 0$ if $m < c$, else $\min(N, \lfloor N \min(m, M)/M \rfloor)$; $\text{exercisable} =$

$\max(0, \text{vested} - \text{exercised})$.

Fund metrics. $\text{DPI} = D/PI$, $\text{RVPI} = \text{NAV}/PI$, $\text{TVPI} = (D + \text{NAV})/PI$, $\text{MOIC} = (\text{real} + \text{unreal})/\text{cost}$.

IRR. solve $\sum_k CF_k/(1+x)^{t_k} = 0$ (Newton–Raphson + bisection).

Cosine similarity. $\text{sim}(u, v) = (u \cdot v)/(\|u\|\|v\|)$.

17 Appendix B — Data Model & API Surface (selected)

Engines. `option_grants`, `grant_exercises`, `option_pools` (share plans); `loans`, `loan_payments`, `loan_covenants` (credit); `valuations_409a` (OPM inputs + derived FMV); `funds`, `fund_lps`, `investments`, capital calls/distributions (Fund OS).

Platform. `investors` (47,275), `contacts`, `founder_submissions`, `news_articles`, `profiles`, `audit_events`, `billing_*`.

REST (selected). `/api/valuations-409a/[id]`; `/api/loans/[id]` (+ `/payments`, `/covenants`); `/api/share-plans/[id]` (+ `/exercises`, `/pool`); `/api/billing/{checkout,portal,webhook}`; `/api/assistant`; `/api/admin/audit`.

Figures marked [FIGURE n] are to be inserted from docs/whitepaper-assets/; testimonials in §14 are placeholders pending named, consented sources; all counts in §13 reflect the production database at the time of writing.

